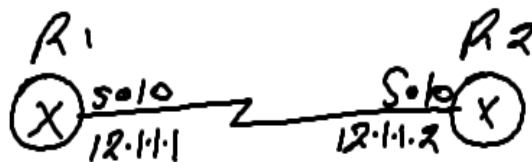


CCNP Route: 300-101

PPP: Point to Point Protocol

- it is open standard protocol
- it is layer 2 protocol
- it can encapsulate any routed protocol. (IPv4, ipv6, IPX, APPLE Talk)
- it support authentication but other layer 2 protocol doesn't support
- it can aggregate multiple serial link between two router
- it also support header compression
- it can use different subnet ip on single segment.
- it support one way authentication and two way authentication
- in ppp we can define static-default route dynamically

how to configure ppp



- r1(config-if)#encapsulation ppp
- r#debug ppp negotiation
- r2(config-if)#encapsulation ppp
- r2#debug ppp negotiation
- r#show ip route

Phases in PPP

1 Down:

- in this phase, ppp is down, this message is seen after the link and ppp are completely down.

2 Establishment phase:

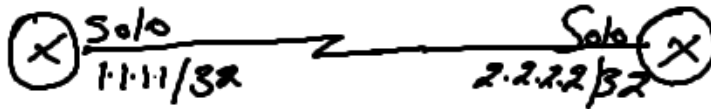
- it performed by LCP protocol
- it will establish link connectivity and will perform authentication
- when device receive an indication that the physical layer is up and ready to be used.

3 Authentication phase:

- PPP Authentication phase is option.
- it occurs when PPP authentication is configured on link, then it will come in authentication phase

4 Network control Phase/UP :

- it performed by NCP protocol
- it will exchange ip address of interfaces between two router (IPCP Protocol)

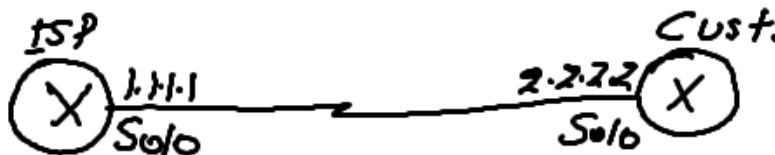


- r1(config)#interface loopback 1
- r1(config-if)# ip address 1.1.1.1 255.255.255.255
- r1(config)#int serial 0/0
- r1(config-if)#ip unnumbered loopback 1
- r2(config)#interface loopback 1
- r2(config-if)# ip address 2.2.2.2 255.255.255.255
- r2(config)#int serial 0/0
- r2(config-if)#ip unnumbered loopback 1
- r#show ip route
- r1(config)#int serial 0/0
- r1(config-if)#no peer neighbor-route (router will not receive route from other router)
- r1#show ip route

Note: CDPCP is responsible to take cdp based packet from one router to another. if CDPCP is disabled then we can't see neighbor info in cdp cmd.

R#clear ppp all (this command can clear process of ppp but it is available in ios 15.0 only)

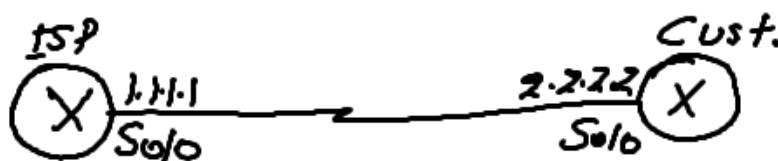
how to configure default-route toward ISP automatically



- cust(config)#int serial 0/0
- cust(config-if)#ppp ipcp route default
- cust#show ip route (sometime route will not come in routing table for that **clear ip route ***)

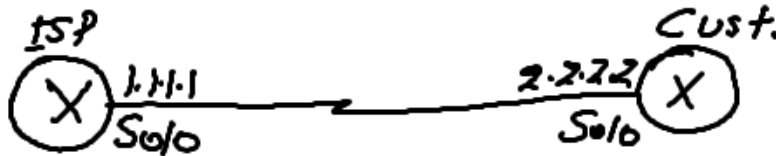
how to configure one-way authentication (PAP)

- in this isp will authenticate to customer router



- isp(config)#username cisco password cisco
- isp(config)#int serial 0/0
- isp(config-if)#ppp authentication pap
- cust(config)#int serial 0/0
- cust(config-if)#ppp pap sent-username cisco password cisco

how to configure two-way authentication (PAP)



- isp(config)#username cisco password cisco
- isp(config)#int serial 0/0
- isp(config-if)#ppp authentication pap
- cust(config)#int serial 0/0
- cust(config-if)#ppp pap sent-username cisco password cisco

- cust(config)#username cisco password cisco
- cust(config)#int serial 0/0
- cust(config-if)#ppp authentication pap
- isp(config)#int serial 0/0
- isp(config-if)#ppp pap sent-username cisco password cisco

how to configure ppp authentication with AAA server.



- isp(config)#AAA new-model
- isp(config)# AAA authentication ppp ccie group radius group tacacs+ local
- isp(config)#username cisco password cisco
- isp(config)#interface serial 0/0
- isp(config-if)#ppp authentication pap ccie
- cust(config)#int serial 0/0
- cust(config-if)#ppp pap sent-username cisco password cisco

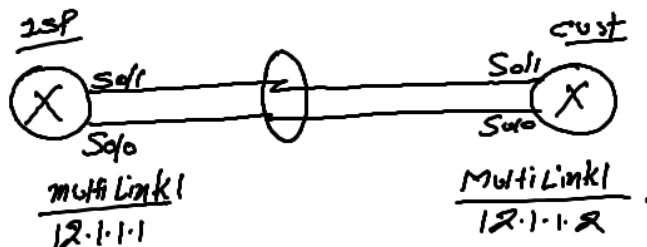
how to connect isp router with radius or tacacs

- `isp(config)#radius-server host 10.10.10.10 key cisco123`
- `isp(config)#tacacs-server host 20.20.20.20`
- `isp(config)#tacacs-server key cisco123`

how to configure chap authentication (two-way)

- `isp(config)#username cisco password cisco`
 - `isp(config-if)#encapsulation ppp`
 - `isp(config-if)#ppp authentication chap`
 - `cust(config-if)#ppp chap hostname cisco`
 - `cust(config-if)#ppp chap password cisco`
-
- `cust(config)#username test password cisco`
 - `cust(config-if)#encapsulation ppp`
 - `cust(config-if)#ppp authentication chap`
 - `isp(config)#int serial 0/0`
 - `isp(config-if)#ppp chap hostname test`
 - `isp(config-if)#ppp chap password cisco`

how to aggregate multiple serial link in ppp



- `isp(config)#interface serial 0/0`
 - `isp(config-if)#encapsulation ppp`
 - `isp(config)#interface serial 0/1`
 - `isp(config-if)#encapsulation ppp`
-
- `isp(config)#interface multilink 1`
 - `isp(config)#int serial 0/0`
 - `isp(config-if)#ppp multilink group 1`
 - `isp(config)# int serial 0/1`
 - `isp(config-if)#ppp multilink group 1`

how to configure authentication in multilink interface

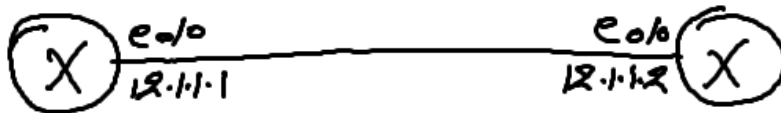
- isp(config)#int multilink 1
- isp(config-if)#ppp authentication chap
- isp(config)#username cisco password cisco
- cust(config)#interface multilink 1
- cust(config-if)#ppp chap hostname cisco
- cust(config-if)#ppp chap password cisco

how to give default route on multilink interface

- cust(config)#interface multilink 1
- cust(config-if)#ppp ipcp route default
- cust#show ip route

PPPoE

- It stand for PPP over ethernet
- it wil add ppp information (8byte) between ethernet header and ip header
- it will not remove ethernet header



PPPoE server configuration

- r1(config)#interface virtual-temple 1
- r1(config-if)#encapsulation ppp (default is ppp)
- r1(config-if)#ip address 12.1.1.1 255.255.255.0 (we can call loopback address)
- r1(config)#bba-group pppoe test
- r1(config-bba-group)#virtual-templete 1
- r1(config)#interface ethernet 0/0
- r1(config-if)#pppoe enable group test

PPPoE client Configuration

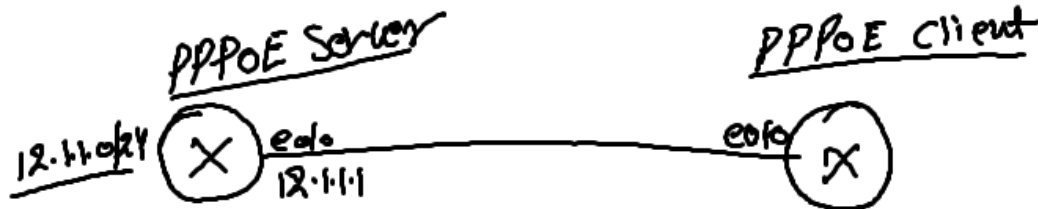
- r2(config)#interface dialer 10
- r2(config-if)#encapsulation ppp
- r2(config-if)#ip address 12.1.1.2 255.255.255.0 (we can call loopback address)
- r2(config-if)#dialer pool 100
- r2(config)#interface ethernet 0/0
- r2(config-if)#pppoe-client dial-pool-number 100

how to configure authentication in PPPoE

- r1(config)#interface virtual-templete 1
- r1(config-if)#ppp authentication chap
- r1(config)#username cisco password cisco

- r2(config)#interface dialer 10
- r2(config-if)#ppp chap hostname cisco
- r2(config-if)#ppp chap password cisco

how to configure dynamic ip address to pppoe client from dhcp server



- r1(config)#ip dhcp pool ccie
- r1(dhcp-config)#network 12.1.1.0 /24
- r1(dhcp-config)#default-router 12.1.1.1
- r1(config)#ip dhcp excluded-address 12.1.1.1
- r1(config)# interface virtual-template 1
- r1(config-if)#peer default ip address dhcp-pool ccie
- r2(config)#interface dialer 10
- r2(config-if)#encapsulation ppp
- r2(config-if)#ip address negotiated / ip address dhcp
- if client is not receiving ip from dhcp server than **clear ppp all** or flap interface

how to provide dynamic ip address to client from local pool

- r1(config)# ip local pool ccie1 12.1.1.10 12.1.1.10
- r1(config-if)# peer default ip address pool ccie1

PPP Negotiation packets:

1 Configure-Request:

- device will send confreq msg to open a connection to the peer.
- it contain configuration option and values the sender wishes the peer to support.
- if the peer responds with a confrej or confnak msg then the router sends another confreq msg.

2 Configure-Reject:

- if configuration option received in the confreq msg is not acceptable thn device will send confreject

3 Configure-Nak:

- if received configuration option is acceptable but some value is not acceptable thn router transmit a confnak msg

Configure-Ack:

- if all options and values in confreq msg are acceptable then router will transmit confack msg

Terminate-Request:

- this msg is used to initiate and LCP close

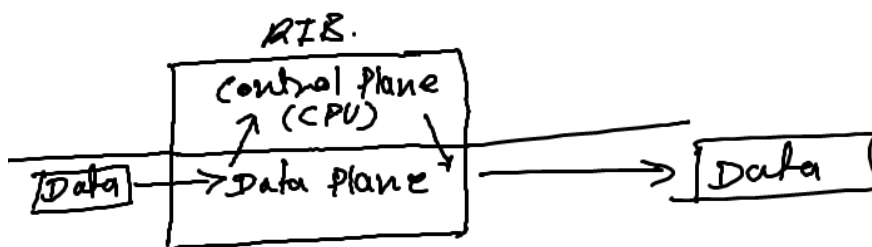
Terminate-Ack:

- this msg is transmitted in response to the terminate-req msg

Types of Layer3 switching

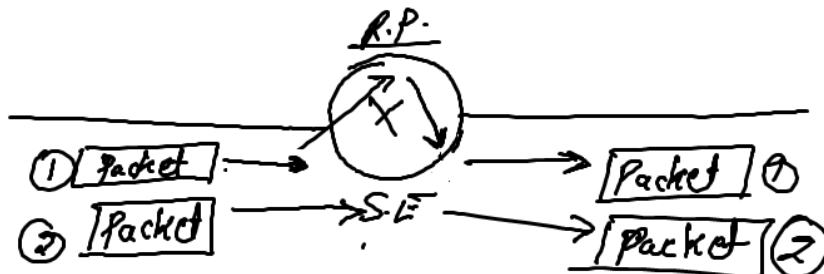
- Process switching
- Fast-switching
- CEF (Cisco Express Forwarding)

Process switching



Fast switching

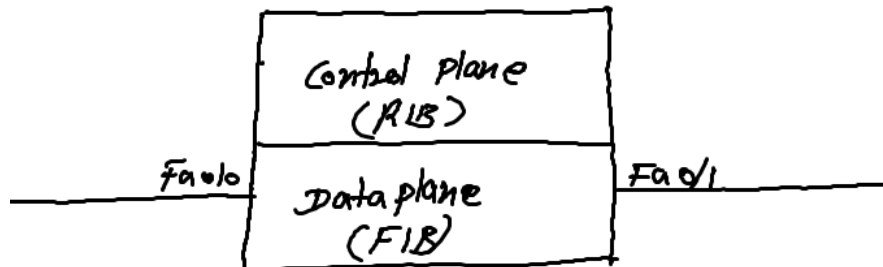
- Route Processor
- Switch engine



CEF (cisco express forwarding)

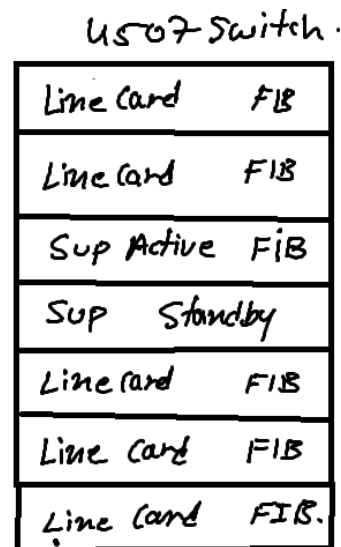
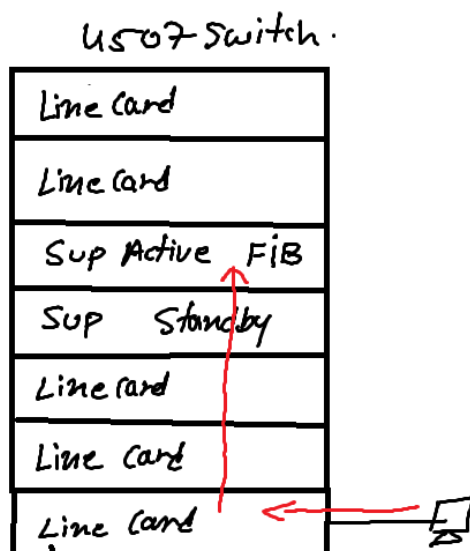
There are two types of table in cef

- FIB: it is replica of RIB, It is present inside data plane only.
- Adjacency



There are two types of CEF

- Centralized CEF
- Decentralized CEF



Type of adjacency table

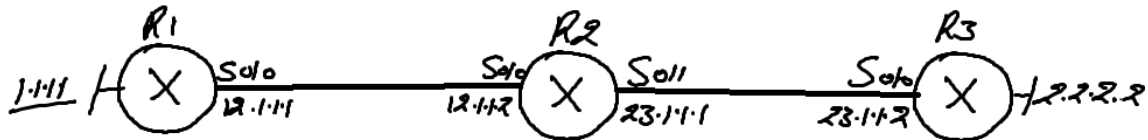
Null adjacency: null adj table will be responsible to handle all those packet which are forwarded towards nulls interface

Drop adjacency: this table is basically responsible to handle all those packet which are encountered with mismatch of encapsulation or crc error.

Discard adjacency: this table is responsible to handle all those packet which are discarded by an acl

Glean adjacency: this table is responsible to have information about all directly connected network and whenever a packet will move to any directly connected network then all those packets will be handled by glean adjacency.

Punt adjacency: this table is responsible to handle those packets which are not processed by CEF and forwarded to control plane to process these packets.



- r1#show ip cef 3.3.3.3
- r#debug ip packet (enable on all routers)
- r2(config)#no ip cef
- r2(config-if)#no ip route-cache

IP addressing version 4, Subnetting and VLSM

ARP Process

IP Routing

- routing is a process or technique to identify the path from one network to another network
- a router must know the following information to route the packets from one network to another network.
 - destination network
 - neighbor router from which it can learn about remote network
 - possible number of routers to reach the destination
 - best route to reach the destination

There are two types of Routing

- Static routing
- Dynamic routing

Static Routing

- Static routing is specified by an administrator in which an administrator manually specifies all routes

Advantage:

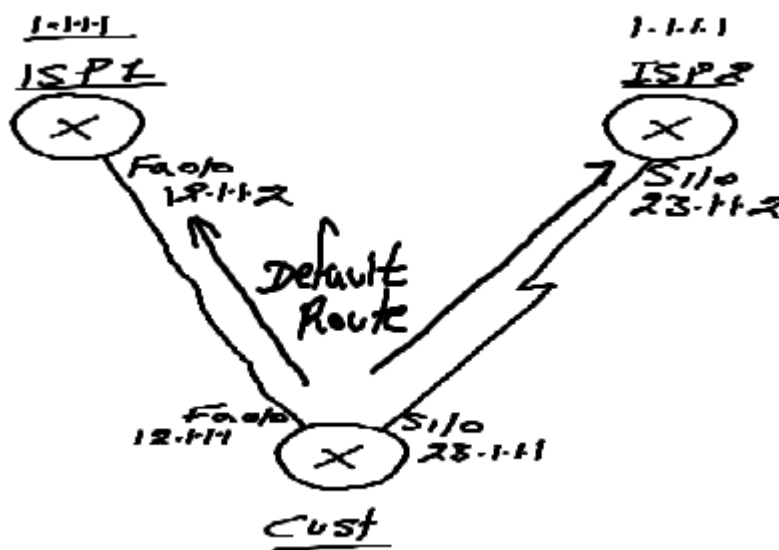
- security
- low CPU utilization
- Efficient in small network
- usually used in small network

Disadvantage:

- Time consuming
- Not good for large network
- burden on an administrator

There are three way to configure static route

- By using next-hop address
- by using exit interface
- by using next-hop and exit interface



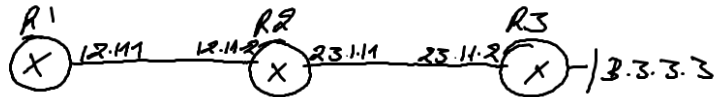
➤ it will install both route into routing table. bcz it will check logest prefix match and prefix match is same. after that it will check a.d. value and a.d. is also same. after that it will metric and

static route metric is always 0 so both route metric is also same.. so it will install both routes into routing table. but it will not perform load-balancing

- it will preffer that path which is installed first into routing table

Floating Static Routing

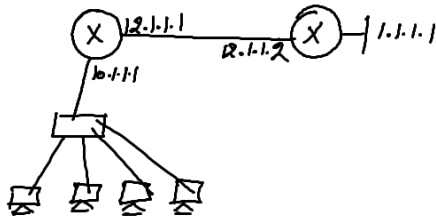
- floating static routing than one route will be save in routing table and second route will be same in nvram if first route goes down than second route will automatically placed into routing table.
- `r(config)#ip route 0.0.0.0 0.0.0.0 12.1.1.1`
- `r(config)#ip route 0.0.0.0 0.0.0.0 23.1.1.1 2`



R1(config)# ip route 3.3.3.0 ~~255.255.255.0~~ 23.1.1.2 (valid or not)

R1(config)# ip route 23.1.1.0 255.255.255.0 12.1.1.2

Q. R1 will communicate with 3.3.3.3 or not?



IP - 10.1.1.0

D.O. - 12.1.1.1

Q ⇒ LAN computer will communicate with loopback 12.1.1.1 or not?

Q ⇒ If it will communicate then why?

EIGRP

- it stand for Enhanced interior gateway routing protocol
- it is open standard protocol.
- it is an IGP Protocol
- it is a hybrid/advance distance routing protocol.
- metric is bandwidth+load+delay+reliability+mtu
- hello timer 5sec and hold timer 15sec.
- hop count bydefault 100 and maximum 255.
- first time full update after that triggered update
- Oit is a PDM Protocol (protocol dependant module).
- it is a classless routing protocol
- it support vlsm
- it support discontinuous network.
- bydefault auto-summary is enabled/ disable in ios version 15.0 and above.
- administrative distance is 90/170/5
- multicast address 224.0.0.10
- protocol number 88.
- there are two version of eigrp v2 and v3
- we can run maximum 30process on a single router
- it support equal load-balancing, bydefault 4, max 16/32.
- it uses dual algorithm for best path selection (diffusing update algorithm)
- it will create neighbor with using primary address.

DUAL (Diffusing update algorithm)

- it is an algorithm used by eigrp for select lowest cost loop free path for each possible destination
- it also provide fast convergence.
- if in topology table multiple path available than it will run dual and it will select best route and put it in routing table
- there are four terms of dual: **1 FD, 2 RD, 3 Successor, 4 Feasible Successor**

FD: Feasible Distance

- total metric from source to destination or you can say that a caculated metric of successor is called FD.

RD: Reported Distance

- A router's FD will be called RD to its neighbor

Successor:

- A successor route is the best route to reach destination network

- a successor route is lowest cost to reach destination
- a successor route will be stored in topology table as well as in routing table.

Feasible Successor

- it is a backup path of successor which is stored in topology table.
- eigrp will keep up to 32 feasible successors in topology table in ios version 15.0

feasibility condition

- $RD < FD$ of current successor

Types of EIGRP packets

- Hello
- Update-RTP
- Query-RTP
- Reply-RTP
- ACK

RTP: Reliable Transport Protocol

- it is used for detecting packet loss and to ensure ordered delivery of the packets

Hello

- hello packet is used for neighbor discovery and for keepalive.

contents of hello

- version
- opcode: hello, ack = 5, update = 1, query = 3, reply = 4
- checksum
- autonomous system number
- eigrp parameters: k values, hold time
- sequence and acknowledged: used for acknowledged packet only (update, query, reply)
- software version: information about the ios version

requirement for neighborship

- A.S. must match.
- K values must match.
- Authentication must match.
- Network id must match.

update:

- update msg can be unicast or multicast
- it uses rtp protocol
- contains (subnets/prefix length, delay, bandwidth, mtu, load, reliability, hop count.)

- update packets are sent between neighbor to build the topology table and routing table.
- if there is static neighborhood between router than update will be unicast.

Ack:

- ack msg always sent as unicast.
- ack is not used in the case of hello.
- it is just for rtp packets

Query:

- query packets are sent by a router when successor route fail and there is no feasible successors in the topology table.
- the router places the route in active state and sent a query to its neighbors for an alternative route.
- query msg are sent as a multicast 224.0.0.10

Reply:

- reply packets are sent in response to query packets, the responding router has an alternative route.
- reply packets are sent as a unicast to the querying router.

stuck in active

- when a router notices that a route failure and there is no feasible successor that route move in active state from passive,
- that router sends a query msg to its neighbor and it will wait for 3min for reply.
- if router does not receive a reply within active timer, that route will be considered stuck-in-active state
- if router not receiving reply within 3min router will flap the neighborhood with its neighbor
- router(config-router)#timers active-time (value)

solution of stuck in active

Disable active timers

Stub router

- router(config-router)#eigrp stub

Summarization

types of table in eigrp

- Neighbor table
- Topology table
- Routing Table

Neighbor Table

- it contains information about its neighbor
- `r#show ip eigrp neighbor`

Topology Table

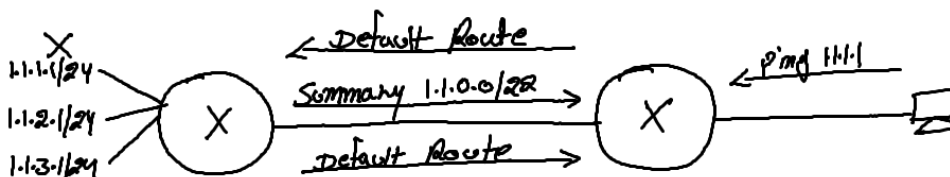
- it contains all destination advertise routes by neighbor routers which come from the nbr routing table
- `r#show ip eigrp topology`
- `r#show ip eigrp topology all-links`

Routing Table

- it contain the best route to each remote network and this path called successor
- it will contain three types of route: internal, external, summary
- `r#show ip route`
- `r#show ip route 1.1.1.0`
- `r#show ip route eigrp`
- `r#show ip route eigrp 100`

Null 0:

- it is a loop avoidance mechanism entry stored in routing table,
- null0 will be created only in case of summarization (auto & manual)
- if any traffic goes towards null0 then it will drop by eigrp



Metric Value: it use composite metric for route selection

- $k1 \cdot 1$ = Bandwidth (measured in kilobytes per second)
- $k2 \cdot 0$ = Load (numbers from 1 to 255)
- $k3 \cdot 1$ = Delay (measured in 10's of microsecond)
- $k4 \cdot 0$ = reliability (numbers from 1 to 255, 255 is most reliable)
- $k5 \cdot 0$ = MTU

serial: bandwidth 1544kbps, delay 20,000 microseconds

fastethernet: bandwidth 100000kbps, delay 1000 microsecond

loopback: delay 5000 microsecond.

how to change metric weights

- R(config-router)#metric weights 0 1 1 1 1 0

Passive Interface

- router(config-router)#passive-interface default
- router(config-router)#passive-interface ethernet 0/0

Unequal-cost Load-Balancing

- eigrp automatically supports load balancing over four equal-cost routes.
- eigrp support up to 32 equal-cost path with ios version 15.0, in previous ios 16path

Varriance: it will multiply with successor when dual algorithm run on topology table

(default1, 1to128).

router(config-router)#varriance (1 to 128)

Offset-list

- it is used for equal cost load-balancing or path manipulation
- router(config)#access-list 1 permit 1.1.1.0 0.0.0.255
- router(config-router)#offset-list 1 in 100 (on all interface)
- router(config-router)#offset-list 1 in 100 ethernet 0/0 (for spacific interface)
- router#show ip protocol (to check offset list).

how to change maxium path

- bydefault eigrp support equal cost load balancing in 4paths
- we can change it up to 32 in ios version 15.0 and above
- router(config-router)#maximum-path 8 (1 to 32)

MD-5 Authentication

- r1(config)#key chain cisco (key chain name can be different)
- r1(config-keychain)#key 1
- r1(config-keychain-key)#key-string cisco1
- r1(config-if)#ip authentication mode eigrp 100 md5
- r1(config-if)#ip authentication key-chain eigrp 100 cisco
- r1#show key chain

MD-5 Authentication with multiple key

- r(config)#key chain cisco
- r(config-keychain)#key 1
- r(config-keychain-key)#key-string cisco1
- r(config-keychain-key)#send-lifetime 10:00:00 16 dec 2015 10:00:00 17 dec 2015

- r(config-keychain-key)#accept-lifetime 10:00:00 16 dec 2015 10:00:00 17 dec 2015
- key 2
- send-lifetime 09:59:00 17 dec 2015 10:00:00 18 dec 2015

interface link working in which A.S

- r#show ip eigrp interface

how to check hello timer on any interface

- r#show ip eigrp interface detail

how to change hello timer and holddown timer

- router(config-if)# ip hello-interval eigrp 100 10
- router(config-if)#ip hold-timer eighrp 100 30

advertise route with redistribution command

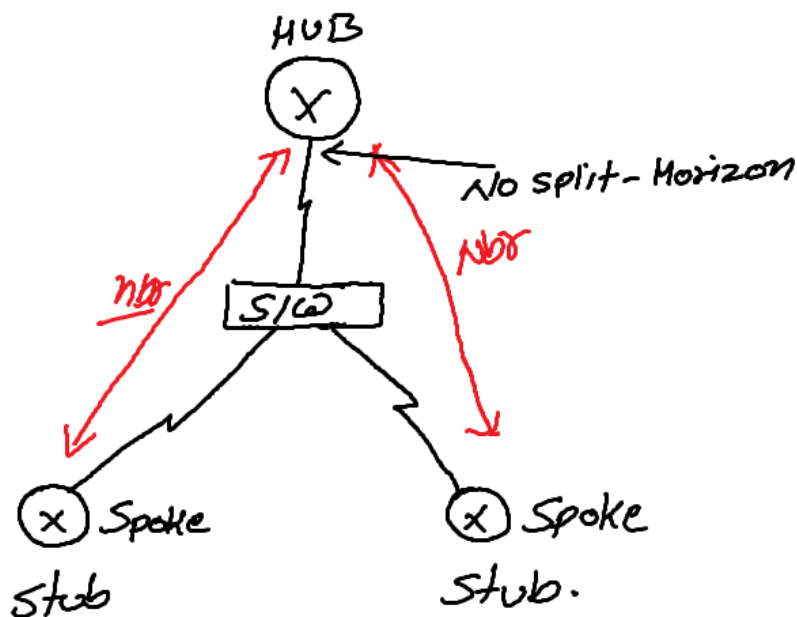
- it will show as a external route in routing table
- router(config-router)#redistribute connected

how to change max hop count

- r(config-if)#metric maximum-hops 1

split-horizon

- it is a loop avoidance technique, by default it is enable on interface
- r(config-if)# No ip split-horizon eigrp 100
- r(config-router)# eigrp stub



Route Filtering

route filtering is used to filter the unused routes of a router. it is used to reduce the size of routing table. we can perform route filtering by using 4 methods.

- Standard ACL
- Extended ACL
- Route-map
- Prefix-list

Route Filtering by using standard Acl

- r(config)#access-list 1 deny 3.3.3.0 0.0.0.255
- r(config)#access-list 2 permit any
- r(config-router)#distribut-list 1 in fa 0/0
- r(config-router)#distribut-list 1 in
- r#show ip protocol

route filtering by using extended Acl

- source: next hop address from which router receiving the routes
- destination: the routes which we want to filter
- r(config)#access-list 101 deny ip host 12.1.1.2 3.3.3.0 0.0.0.255
- r(config)#access-list 101 permit ip any any
- r(config-router)#distribut-list 101 in
- r#show ip protocol
- r#show access-list 101

Route filtering by using Route-Map

- r(config)#access-list 1 permit 3.3.3.0 0.0.0.255
- r(config)#route-map cisco deny 10
- r(config-route-map)#match ip address 1
- r(config)#route-map cisco permit 20
- r(config-router)#distribute-list route-map cisco in
- r#show route-map

Route filtering by metric

- r(config)#route-map cisco deny 10
- r(config-route-map)#match metric 1000
- r(config)#route-map cisco permit 11
- r(config-router)#distribut-list route-map cisco in

filter route with using metric between 100 to 200

200-100 =100

100/2=50

100+50=150 (150 +- 50)

- router(config)#route-map cisco deny 10
- router(config-route-map)#match metric 150+-50
- router(config)#route-map cisco permit 11
- router(config-router)#distribute-list route-map cisco

Route filtering by using prefix-list

- any= 0.0.0.0 /0
- ge= greater thn
- le= less thn

- r(config)#ip prefix-list 1 deny 1.1.0.0/16 ge 24 le 24
- r(config)#ip prefix-list 1 permit 0.0.0.0/0 le 32
- r(config-router)#distribute-list prefix 1 in
- r#show ip prefix-list 1

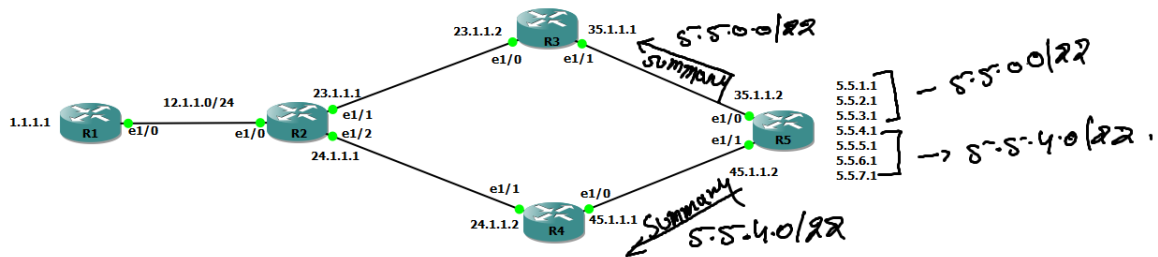
Route filtering by using A.D. 255

- r(config)#access-list 1 permit 1.1.1.0 0.0.0.255
- r(config-router)#distance 255 12.1.1.2 0.0.0.0 1 (for specific route from 12.1.1.2)
- or
- r(config-router)#distance 255 12.1.1.2 0.0.0.0 (for all route from 12.1.1.2)

how to filter internal/external route

- r(config-router)#distance eigrp 90 255.
- r(config-router)#distance eigrp 255 170.

Load-balancing with using summarization



```
r5(config-if)#ip summary-address eigrp 100 5.5.0.0 255.255.252.0 (interface fa 1/0)
```

```
r5(config-if)#ip summary-address eigrp 100 5.5.4.0 255.255.252.0 (interface fa 1/1)
```

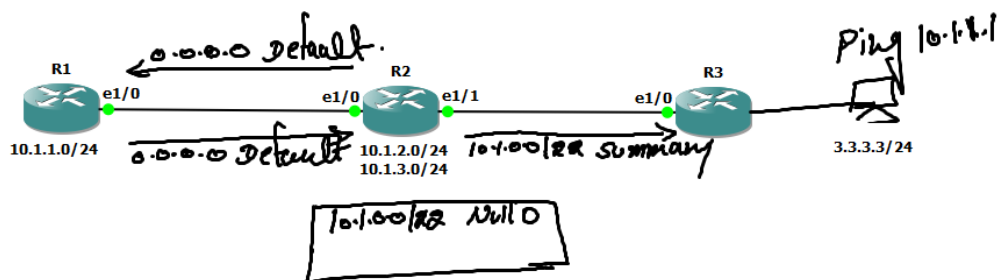
how to configure summarization with leak-map

- r5(config)#access-list 10 permit 5.5.3.1 0.0.0.255
- r5(config)#route-map cisco permit 5
- r5(config-route-map)# match ip address 10
- r5(config-if)#ip summary-address eigrp 100 5.5.0.0 255.255.0.0 leak-map cisco

how to change A.D. value for summary route/ for null

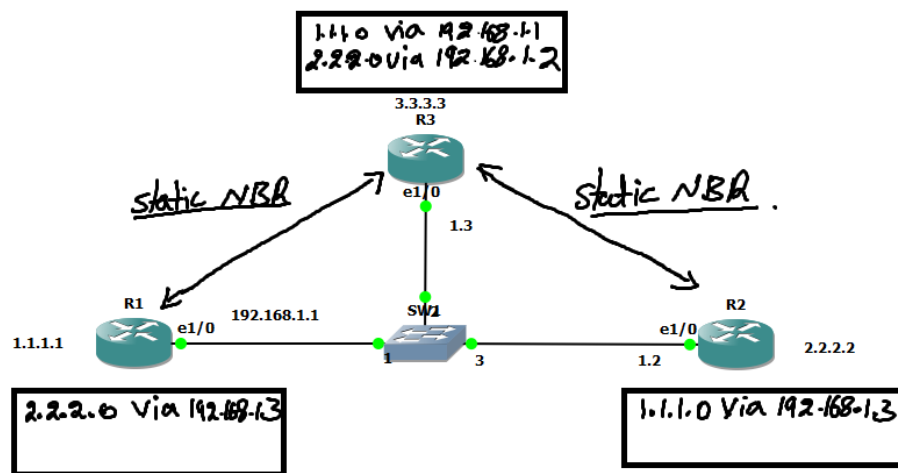
```
r5(config-if)#ip summary-address eigrp 100 5.5.0.0 255.255.0.0 91
```

how to restrict router to install summary route toward null0



```
r5(config-if)#ip summary-address eigrp 100 10.1.0.0 255.255.252.0 255
```

static neighborship



```
r3(config-if)#no ip split-horizon eigrp 100
```

```
r3(config-if)#no ip next-hop-self eigrp 100
```

Propagate/Generate default route in EIGRP

- Redistribut static
- summarization
- ip default-network
- network 0.0.0.0

redistribute static

- router(config)# ip route 0.0.0.0 0.0.0.0 null 0
- router(config-router)# redistribute static

summarization

- router(config-if)# ip summary-address eigrp 100 0.0.0.0 0.0.0.0

IP default-network

- r(config)#interface loopback 1
- r(config-if)#ip address 192.168.101.1 255.255.255.0
- r(config)#router eigrp 100
- r(config-router)#network 192.168.101.0
- r(config)# ip default-network 192.168.101.0

network 0.0.0.0

- router(config)# ip route 0.0.0.0 0.0.0.0 null 0
- router(config-config)#network 0.0.0.0

Redistribute rip/ospf into EIGRP

- router(config-router)#redistribute rip metric 1544 0 255 1 1500
- router(config-router)#redistribute ospf metric 1544 0 255 1 1500

- or
- router(config-router)#redistribute rip
- router(config-router)#redistribute ospf 100
- router(config-router)#default-metric 1544 0 255 1 1500

advance redistribution in eigrp

- router(config)# route-map tag1 permit 10
- router(config-route-map)#set tag 600
- router(config-router)# redistribute rip metric 1544 0 255 1 1500 route-map tag1

- router(config)#route-map tag2 permit 10
- router(config-route-map)#set tag 700
- router(config-router)# distribute ospf 100 metric 1544 0 255 1 1500 route-map tag2

- router(config)#route-map tag3 permit 10
- router(config-route-map)#set tag 800
- router(config-router)#redistribute eigrp 200 route-map tag3

- router#show ip route 5.5.5.0
- router#show ip eigrp topology 5.5.5.0 255.255.255.0
- router#show ip eigrp topology

Route filtering with tag value

- router(config)#route-map filter deny 10
- router(config-route-map)#match tag 700
- router(config)#route-map filter permit 11
- router(config-router)#distribute-list route-map filter in

filter external/internal route with using route-map

- router(config)#route-map filter1 deny 10
- router(config-route-map)#match route-type external/internal
- router(config)#route-map filter1 permit 11
- router(config-router)#distribute-list route-map filter1 in

filter route from specific protocol

- router(config)#route-map ext deny 10
- router(config-route-map)#match source-protocol rip/eigrp200/ospf 100/static/connected
- router(config)#route-map ext permit 11
- router(config-router)#distribute-list route-map ext in

Router-id in eigrp

- external route will carry router-id of originating router.

- router#show ip eigrp topology 5.5.5.0 255.255.255.0

how to electe Router-ID in eigrp

- router(config-router)# eigrp router-id 10.10.10.10
- 1 Highest IP address of loopback interface
- 2 Highest IP address of physical interface

EIGRP Stub Routing

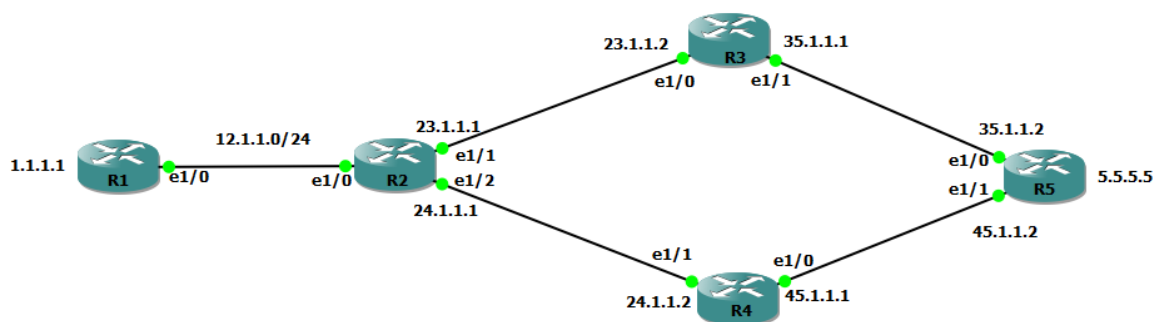
- r(config-router)# eigrp stub (it will forward connected and summary route only)
- r(config-router)#eigrp stub receive only (it will not forward any route)
- r(config-router)#eigrp stub connected (it will forward only connected route)
- r(config-router)# eigrp stub summary (it will forward only summary route)
- r(config-router)# eigrp stub redistributed (it will forward only redistributed route)

Policy based routing

- Policy based routing can be used to change the next hop ip address. this can be usefull to over rule your routing table for certain traffic types.
- router(config)#access-list 101 permit ip any 3.3.3.0 0.0.0.255
- router(config)#route-map cisco permit 10
- router(config-route-map)#match ip address 101
- router(config-route-map)# set ip next-hop 13.1.1.2
- router(config-if)#ip policy route-map cisco
- router#debug ip policy

IP SLA (IP Service level agreement)

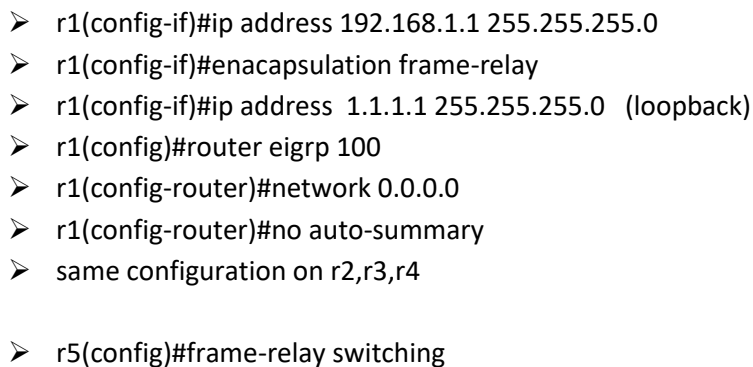
it is a track object. this feature allow you to make sure that next hop is reachable befor that the routing table is used.



- router(config)#access-list 101 permit ip any 5.5.5.0 0.0.0.255
- router(config)#ip sla 10
- router(config-ip-sla)#icmp-echo 45.1.1.2 source-ip 24.1.1.1
- router(config-ip-sla-echo)#frequency 5 (default 60sec)
- router(config)#ip sla schedule 10 start-time now life forever

- ### how to change bandwidth usage by eigrp (by default uses 50%)

- ## Frame-relay fullMesh



- r5(config)#interface serial 0/0
- r5(config-if)#encapsulation frame-relay
- r5(config-if)#frame-relay intf-type dce
- r5(config-if)#clock rate 64000
- r5(config-if)#frame-relay route 102 interface serial 0/1 201
- r5(config-if)#frame-relay route 103 interface serial 0/2 301
- r5(config-if)#frame-relay route 104 interface serial 0/3 401

- r5(config)#interface serial 0/1
- r5(config)#encapsulation frame-relay
- r5(config-if)#frame-relay intf-type dce
- r5(config-if)#clock rate 64000
- r5(config-if)#frame-relay route 201 interface serial 0/0 102
- r5(config-if)#frame-relay route 203 interface serial 0/2 302
- r5(config-if)#frame-relay route 204 interface serial 0/3 402

- r5(config)#interface serial 0/2
- r5(config)#encapsulation frame-relay
- r5(config-if)#frame-relay intf-type dce
- r5(config-if)#clock rate 64000
- r5(config-if)#frame-relay route 301 interface serial 0/0 103
- r5(config-if)#frame-relay route 302 interface serial 0/1 203
- r5(config-if)#frame-relay route 304 interface serial 0/3 403

- r5(config)#interface serial 0/3
- r5(config)#encapsulation frame-relay
- r5(config-if)#frame-relay intf-type dce
- r5(config-if)#clock rate 64000
- r5(config-if)#frame-relay route 401 interface serial 0/0 104
- r5(config-if)#frame-relay route 402 interface serial 0/1 204
- r5(config-if)#frame-relay route 403 interface serial 0/2 304
- neighborship will established between neighbor
- r5#show frame-relay route
- r5#show frame-relay pvc

disable inverse arp

- router(config-if)#no frame-relay inverse-arp (disable on all router)
- router#clear frame-relay inarp (to clear inverse arp-cache)

how to configure manual mapping

- router1(config-if)#frame-relay map ip 192.168.1.2 102 broadcast

hub and spoke topology

- router(config)#interface serial 0.0.1 point-to-point
- router(config-subif)#ip address 12.1.1.1 255.255.255.0
- router(config-subif)#frame-relay interface-dlci owndlci

Named mode Eigrp

- **Supports both IPv4 and IPv6 EIGRP** Can be in different ASes
- **Better config hierarchy**
- **Wide Metrics, IPv6 VRF Lite, etc**

- router(config)#router eigrp ccie
- router(config--router)# address-family ipv4 autonomous-system 100
- router(config-router-af)#network 12.1.1.0 0.0.0.255
- router#show ip protocol
- router#show ip topology
- router#show ip route

- router(config-router)#address-family ipv4 autonomous-system 100
- router(config-router-af)# af-interface default/ ethernet 1/0
- router(config-router-af-interface)#passive-interface
- router(config-router-af-interface)#hello-interval 7
- router(config-router-af-interface)#hold-interval 15
- router(config-router-af-interface)#authentication mode md5
- router(config-router-af-interface)#authentication key-chain cisco123
- router(config-router-af-interface)#summary-address 10.1.0.0 255.255.252.0

- router(config-router)#address-family ipv4 autonomous-system 100
- router(config-router-af)# topology base
- router(config-router-af-topology)#timers active-time 4min/disable
- router(config-router-af-topology)# metric maximum-hop 2
- router(config-router-af-topology)#maximum-path 32
- router(config-router-af-topology)#variance 2 (1 to 128)
- router(config-router-af-topology)#auto-summary
- router(config-router-af-topology)#offset-list 1 in 1000
- router(config-router-af-topology)#distribute-list acl/prefix/route-map

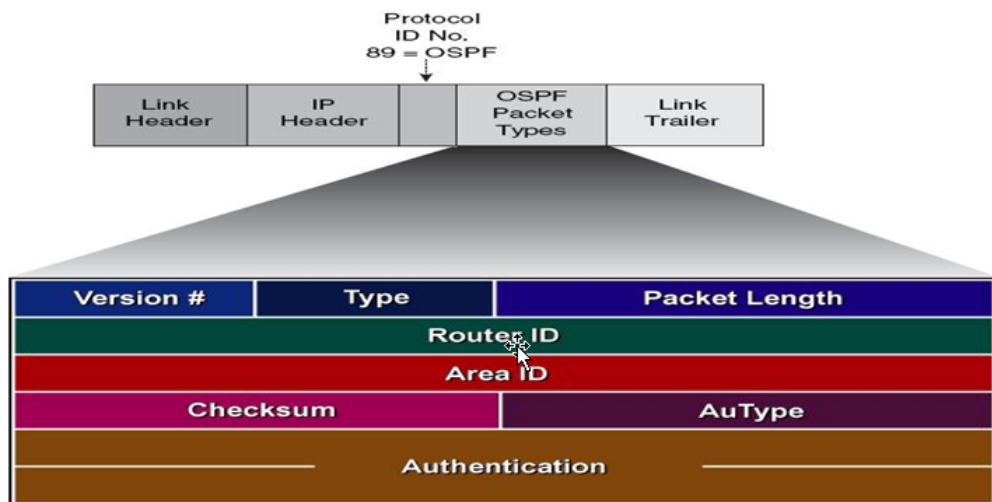
OSPF Open Shortest Path First

- it is a open standard protocol.
- It is an IGP Protocol.
- it is a link state routing protocol.
- first time full update after that partial update.
- hop count limit is unlimited.
- metric= cost
- it uses protocol number 89.
- Default administrative distance is 110.
- hello timer 10sec, hold time 40.
- multicast address 224.0.0.5(for all router), 224.0.0.6(for DRrouter).

- it is a classless routing protocol.
- by default auto-summarization is disabled.
- maximum path by default 4 (ios 15.0 = 32).
- it sends an update with a sequence number starting from 0x80000001 to 0xffffffff

Types of packet in ospf

- Type1 hello
- Type2 DBD database description
- Type3 LSR Link state request
- Type4 LSU Link state update (retransmission time out 5sec)
- Type5 LS Ack link state acknowledge



Note: OSPF header size is 32bytes but always remember that first 24bytes are always common in all packet

OSPF Design:

- it reduces the memory and CPU utilization of router
- it is easy to solve issues in smaller domain that is area

Area in ospf

- Area is a logical group of devices within a single administration.
- Cisco recommends that in a single area should not have more than 30 routers
- area-id is a 32bit long ID.
- Area-id can be represented in decimal number or IPv4 format

Note: inter area route can not move from one area to another area without backbone area, but external route can move from one area to another area without backbone area.

Types of Router in OSPF

- Backbone router
- internal router
- Area Border Router
- Autonomous System Boundary router

Backbone router

- a router has all interfaces in Area 0
- we can call it transit area

internal router

- A router which has all interfaces in regular Area

ABR Area Border Router

- a router which has at least one interface in backbone area and one in regular area
- we create ABR for device database in different area
- An ABR will keep multiple Area's database in memory
- it can send routing information to different area but not send database

ASBR autonomous system border router

- it is used to connect different routing protocol with ospf or redistribute rip,eigrp into ospf
- it is used to exchange routing information in ospf and different protocol

Requirement to form ospf neighborship

- area-id must match
- timers must match
- authentication must match
- subnetmask must match
- stub flag must match
- mtu size must match
- router-id must not match
- network type should match

contents of hello msg

- ospf version
- area id
- router id
- network mask
- hello interval
- checksum
- dead interval
- stub flag

- priority
- authentication type
- authentication
- designated router
- backup designated router
- neighbor router id
- neighbor router priority

Router-id

- router id is a unique ID in a area to identify the database on every router

how to set router id manually

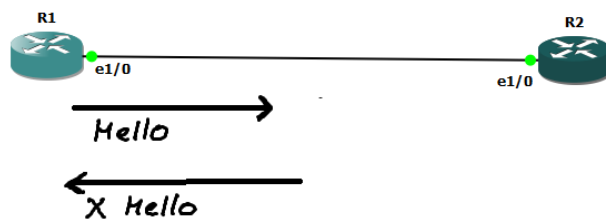
- router(config-router)#router-id 1.1.1.1

how to set router-id automatically

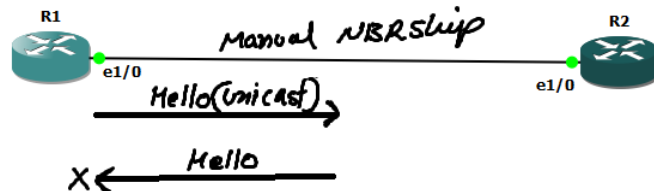
- highest ip address of loopback interface
- highest ip address of physical interface

ospf neighborship states

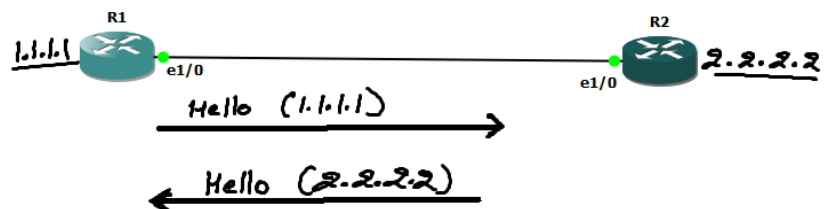
1 Down State



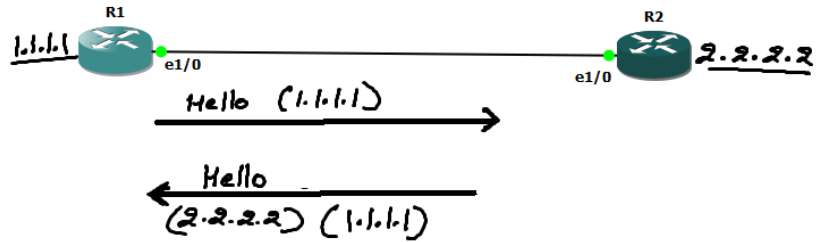
2 Attempt



3 init state



4 Two Way



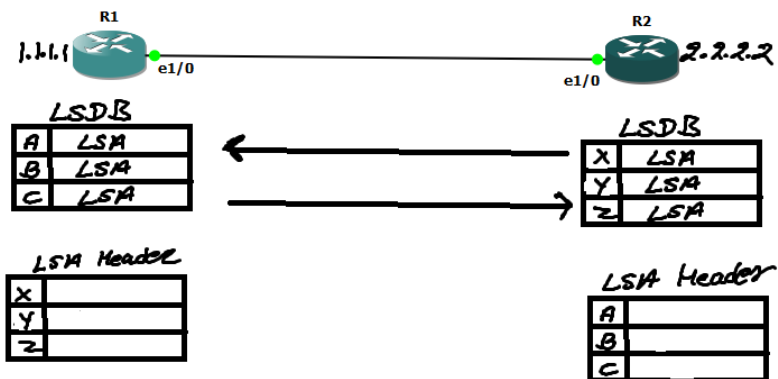
5 Ex-start



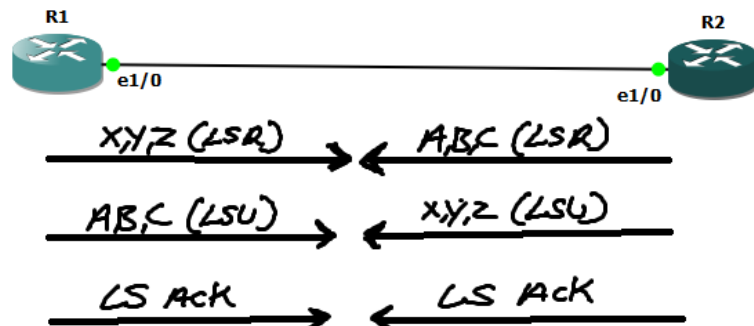
⇒ router will negotiate the seq no. of DBD and mtu size

⇒ in this state router will exchange empty DBD for master and slave election

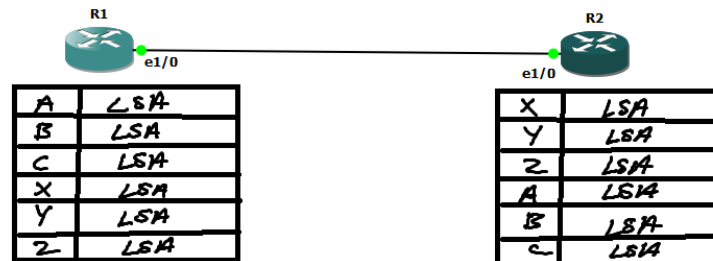
6 Ex-change state



7 Loading state



8 Full state



Routers have synchronized database successfully

Formula to calculate total number of adjacency

when we don't have a DR and BDR $N(n-1)/2$

when we have both DR and BDR $N \times 2 - 3$

when we have only DR $(N-1)$

LSA: Link State Advertisement

A router will always keep its route information in its database in the form of LSA, when any update will go from one device to another device it will go in the form of LSA

Types of LSA

- Type 1 LSA: Router LSA
- Type 2 LSA: Network LSA
- Type 3 LSA: Summary LSA
- Type 4 LSA: ASBR Summary LSA
- Type 5 LSA: External LSA
- Type 6 LSA: Group Membership LSA
- Type 7 LSA: NSSA LSA
- Type 8 LSA:
- Type 9 LSA:
- Type 10 LSA:
- Type 11 LSA:

Type 1 LSA: Router LSA

- It is generated by every router within an area to represent itself, it will not cross its own area
- within an area all routers have common 'Router LSA' information,
- one router LSA can keep multiple links information
- after every 30min router will flood LSA, if any router receives higher sequence no. LSA than router will execute SPF algorithm
- in router LSA link-id and advertising router-id will be same
- router LSA contain some special bits like 'V' end point of virtual link, 'E' it is ASBR, 'B' it is ABR.
- link-id= router id, adv router-id= router-id

Type 2 LSA: Network LSA

- It is generated by DR router within a segment to represent itself
- it is having information that how many devices are connected in that multi-access link network in the form of router-id. it will also having the information of DR router's ip address.
- it send to all attached router between the same area
- link-id= ip address of DR, adv router-id= router id of DR

Type 3 LSA: summary LSA

- It is generated by ABR router for sending routing information from one area to another area

- it contain information about all inter area routes
- link-id=network id (prefix id), adv. router-id= router id of ABR

Type4 LSA: ASBR summary LSA

- it is generated by ABR
- describe that who is doing redistribution means ASBR information
- cost to reach an ASBR from ABR
- Link-id= router-id of ASBR, Adv. router-id= router-id of ABR.

Type5 LSA: External LSA

- generated by ASBR
- this LSA will keep the information of all routes of other routing domain
- link-id= external routes, adv. router-id= router-id of ASBR.

Type7 LSA: NSSA External LSA

- It is generated by ASBR within NSSA Area
- it remain within the NSSA Area
- LSA7 gets translate back into type5 LSA by the NSSA ABR router
- link-id= external routes, adv. router-id= router-id of ASBR

how to enable ospf on interface

- router(config)#router ospf 100
- router(config-router)#network 12.1.1.0 0.0.0.255 area 0
- router(config-if)#ip ospf 100 area 1

how to elect DR/BDR on point to point link

- router(config-if)#ip ospf network broadcast

how to avoid DR/BDR election on multi-access link

- router(config-if)#ip ospf network point-to-point

how to change priority of interface

- router(config-if)#ip ospf priority 2 (0-255)

how to advertise loopback on there own prefix length

- router(config-if)#ip ospf network point-to-point

how to check ABR router

- router#show ip ospf border-router

how to check which interface in which area

- router#show ip ospf interface brief

how to check detail of all interface

- router#show ip ospf interface

how to configure virtual-link

- router(config-router)#area 1 virtual-link 3.3.3.3

redistribute eigrp into ospf

- router(config-router)#redistribute eigrp 100 subnets

redistribute rip into ospf

- router(config-router)#redistribute rip subnets

Note: all external route will come as E2 route in ospf and it is larger route than any other path. it will never show total cost of path (cost20)

advance redistribution in ospf (change seed-metric)

- router(config-router)#redistribution eigrp 100 subnets metric 30

Note: if cost to reach asbr is same than we can change seed-metric

how to advertise E1 route during redistribution

- router(config-router)#redistribution eigrp 100 subnets metric-type 1

how to add tag during redistribution

- router(config-router)#redistribution eigrp 100 subnet metric-type 1 tag 11

condition redistribution with using route-map (redistribute specific route)

- router(config)#access-list 1 permit 60.1.1.0 0.0.0.255
- router(config)#access-list 1 permit 60.1.2.0 0.0.0.255
- router(config)#route-map cisco permit 10
- router(config-route-map)#match ip address 1
- router(config-router)#redistribution eigrp 100 subnet metric metric-type 1 tag 11 route-map cisco

redistribute ospf into eigrp for specific route

- router(config)#access-list 1 permit 10.1.1.0 0.0.0.255
- router(config)#route-map cisco1 permit 10
- router(config-route-map)#match ip address 1
- router(config)#router eigrp 100
- router(config-router)#redistribution ospf 1 metric 1544 0 255 1 1500 route-map cisco1

Type of Area

- Standard Area
- Stub Area
- Totally Stub Area
- NSSA Area
- Totally NSSA Area

Note: we used area type to filter LSA

Standard Area:

- by default all areas are standard area

Stub Area

- it is used to filter Type5 LSA and Type4 LSA or to filter external routes
- after filtering type5 and type4 LSA It automatically generate default route
- default route is advertise by ABR
- stub are run on both sides means ABR and all other router of this area.
- we cant configure area0 as stub area
- stub area not allow virtual-link
- stub area is not allowed ASBR router so we cant configure stub area near ASBR router.
- we cant make stub to that area which is attached with ASBR bcz it cant do redistribution .
- router(config-router)#area 1 stub

Totally stub:

- it filter LSA3,4,5 or external as well as inter-area routes
- It automatically generate Default route
- implement only on ABR router
- router(config-router)#area 1 stub no-summary

NSSA Area

- it stand for not-so-stub-area
- it is used to allow an asbr to send external routes through stub area with using type7 LSA.
- filter LSA4,5 but redistribution is allowed
- ASBR router will generate type7 LSA after that ABR will convert type 7 lsa back to type5 LSA.
- default route will be manually configured
- run on both side like stub router
- router(config-router)#area 1 nssa
- router(config-router)#area 1 nssa default-information originate

Totally NSSA

- it filter LSA3,4,5
- Default route will automatically generated
- implement only on ABR router
- allow redistribution
- default route originate by ABR
- router(config-router)# area 1 nssa no-summary

propagate default-route in ospf with using default-information originate command

- router(config-router)#default-information originate / metric-type 1

it will propagate default route if default route is present in its routing table

- router(config-router)# default-information originate metric-type 1 always

Route filtering

To filter routes we can use distribute-list. under distribute-list we can call various components

- Standard ACL
- Extended ACL
- Prefix-list
- Route-MAP
- Area Filter List

Note: we can't configure Distribute-list in outbound direction in ospf except ASBR

Standard ACL

- router(config)#access-list 10 deny 100.1.1.0 0.0.0.255
- router(config)#access-list 10 permit any
- router(config-router)#Distribute-list 10 in

Extended ACL

- router(config)# access-list 101 deny ip host 12.1.1.2 172.16.111.0 0.0.0.255
- router(config)#access-list 101 permit ip any any
- router(config-router)#distribute-list 101 in

Prefix-list

- router(config)#ip prefix-list ccie deny 172.16.0.0/16 ge 24 le 24
- router(config)#ip prefix-list ccie permit 0.0.0.0/0 le 32
- router(config-router)#distribute-list prefix ccie in

Route-Map with specific metric

- router(config)#route-map test deny 10
- router(config-route-map)#match metric 20

- router(config)#route-map test permit 11
- router(config-router)#distribute-list route-map test in

route-map

- router(config)#access-list 1 permit 10.1.1.0 0.0.0.255
- router(config)#route-map cisco deny 10
- router(config-route-map)# match ip address 1
- router(config-router)#distribute-list route-map cisco in

Filter Route on ASBR for outbound direction

- router(config)#access-list 1 deny 100.1.1.0 0.0.0.255
- router(config)#access-list 1 permit any
- router(config-router)#distribute-list 1 out

Area Filter-list

- In Area filter-list we can use only prefix-list
 - Area filter-list can configure both direction
 - we can filter inter-area routes
 - we can filter type3 LSA for a area
 - we can configure area filter-list on ABR router only
-
- router(config)#ip prefix-list ccie deny 10.1.0.0/16 ge 32 le 32
 - router(config)#ip prefix-list ccie permit 0.0.0.0/0 le 32
 - router(config-router)#area 1 filter-list prefix ccie in
 - or
 - router(config-router)#area 0 filter-list prefix ccie out

summarization

- we can't configure summarization for intra area route
- we can configure summarization only on ABR which is generating LSA3 or on ASBR which is generating LSA5.

Types of summarization

- Internal (On ABR)
- External (On ASBR)

internal summarization

- router(config-router)# area 0 range 10.1.0.0 255.255.252.0 (routes of area 0 on R7)

remove null 0 entry for internal summarization

- router(config-router)#no discard-route internal

external summarization

- router(config-router)# summary-address 1.1.0.0 255.255.252.0

remove null 0 entry for external summarization

- router(config-router)#no discard-route external

note: we can configure external summarization on translater also.

filter routes with using internal summarization

- router(config-router)# area 2 range 1.1.0.0 255.255.252.0 not-advertise

filter routes with using external summarization

- router(config-router)# summary-address 1.1.0.0 255.255.252.0 not-advertise

how to change max-path in ospf

- router(config-router)#maximum-path 10

how to change A.D. value in ospf

- router(config-router)#distance 100

how to change hello-timer in ospf

- router(config-if)#ip ospf 100 hello-interval 5

note: if we changed hello interval than dead timer will automatically get changed. if we changed dead-interval manually than dead will not get change automatically

how to change dead-interval in ospf

- router(config-if)#ip ospf 100 dead-interval 15

how to check hello and dead timer in ospf

- router# show ip ospf interface serial 0/0

note: if hello and dead timers are mismatch than neighborhood will not come up

how to change reference bandwidth in ospf

- router(config-router)#auto-cost reference-bandwidth 1000
- router#show ip protocol

how to check cost of any interface

- router#show ip ospf interface serial 0/0

how to check bandwidth of any interface

- router#show interface serial 0/0

how to change cost on any interface

- We can change path cost for path manipulation
- router(config-if)#ip ospf cost 50

how to set maximum LSA size

- router(config-router)#max-lsa 40 / warning-only

authentication in ospf

- Type 0 - Null
- Type1 - Plain text
- Type2 - MD-5

How to configure plain text authentication

- router(config-if)# ip ospf authentication (to enable plain text authentication)
- router(config-if)#ip ospf authentication-key cisco123
- router#debug ip ospf events

how to configure md-5 authentication

- router(config-if)# ip ospf authentication message-digest (to enable md-5 authentication)
- router(config-if)#ip ospf message-digest-key 1 md5 cisco123
- router#debug ip ospf events

how to set null authentication

- router(config-if)#ip ospf authentication null

note: we can configure plain text and md5 authentication but at a time latest authentication will work

how to configure authentication globally

router(config-router)#area 0 authentication message-digest

router(config-if)#ip ospf message-digest-key 1 md5 cisco123

OSPF Network types

we use network types in ospf bcz it depend upon a media

RFC Specific: non broadcast and point to multipoint

Cisco specific: point to point, point to multipoint non broadcast and broadcast

Fully mesh topologies

- a. Broadcast
- b. Non broadcast

Hub & Spoke

p2mp, p2mp-nb, p2p

1. Non broadcast n/w –

Default n/w type for frame relay (in case of sub-interface multipoint).

Manually specify neihrshp becz it can't make nbr automatically (hello 30 and dead 120)

next hop can't change.

Unicast hello send and Can connect more than 2 router but have no broadcast capability.

It elects Dr and BDR but all OSPF packets are exchange through unicast

Command: Int fa0/0

Ip ospf network non-broadcast

Note: Not mandatory to configure static nbrshp both side. we can configure only one side

2. Broadcast n/w –

Default n/w on fast Ethernet and FDDI

Will elect DR & BDR

Always map with broadcast keyword otherwise can't make nbrship.

We can't able to set nbr manually if we forget to mention broadcast keyword in mapping time.

It use multicast 224.0.0.5 add (with 01005e00005 for all ospf router) 224.0.0.6 for all DR router.

Hello time 10 sec dead 40 sec

To change n/w type

Int fa0/0

#IP OSPF n/w broad cast

Point to multipoint :-

it is not default option but best choice for nbma n/w, if broadcast is enabled otherwise it can't work

Does not select DR and BDR packet are multicast 224.0.0.5

Can't be configured manually assign nbr

Hello time 30 sec dead time 120 sec

/32 will be as entry of all routers as a host.

Point to multipoint non-broadcast:-

Same as point to multipoint but configure with additional non broadcast keyword

No DR/BDR election

Unicast neighbor statement should be configured

The packet is unicast

Hello time 30sec dead time 120sec

Point to point n/w :-

Default for TI, point to point sub interface on frame relay

Hello time 10 sec and dead interval 40.

next hop ip is advertizing router.

No DR and BDR election.

It use multicast add -224.0.0.5

OSPF ignore subnet mask mismatch on point to point link.

6. Loopback:-

1. It is default n/w type for loopback interface

2. OSPF advertise stub n/w as host router /32

7. virtual-link n/w:-

Virtual link is used for non backbone area that is not directly connected with backbone area (area 0).

We always configure a virtual link over a primary address.

This n/w type suppress hello

Transits area means who work between area 0 and any regular area.then we use virtual-link over transits area.

Area 1 virtual- link 146.146.146.6 hello interval 20

Sh ip os virtual-link

n/w types	Hello times	Dead times	DR & BDR	Auto NBR discovery	Manual nbr define
Broadcast	10	40	Y	Y	X
P2p	10	40	X	Y	X
P2mp	30	120	X	Y	X
NON BROADCAST	30	120	Y	N	Y
P2mp (NB)	30	120	X	N	Y

BGP: Border Gateway Protocol

- it is an EGP Protocol (it can form neighborhood between two different AS's)
- it is a Path Vector routing Protocol (it uses multiple attributes to select best path)
- it is highly scalable than IGP (it can have lakhs of routes in its BGP table)
- it is reliable protocol (it uses TCP)
- it uses tcp port number 179 before establish neighborhood it have to form tcp connection (three way hand shake process).
- only BGP is fully reliable routing protocol.
- it is more secure than any other routing protocol bcz all msgs will travel on TCP protocol.
- it is a classless routing protocol.
- it support manual summarization
- Administrative Distance: IBGP 200, EBGP 20.
- A.S. size is 2bytes(0-65535) and 4bytes(0.0-65535.65535).
- Public A.S= 1 to 64511, Private AS=64512 to 65534.
- BGP doesn't support dynamic neighbor discovery.
- it supports only md5 authentication
- bgp convergence is slow as compare to IGP.

note: we can't configure more than one AS on a single router.

note: BGP sends all type of packets as unicast.

why we use BGP

- IGP's were developed for to handling upto 1000 of routes inside their routing table. therefore can't be implement in very large networks.
- BGP can contain lakh's of routes inside their BGP table or routing table.
- we never use default route in ISP router to other ISP router.
- in IGP's if any link (prefix) goes down thn it effects to complete routing structure(topology). but in BGP, it not affects other routers bgp table.
- when we are service provider or network size is very large.
- when we have multiple connections for internet or multiple service provider.

Types of customers

- single homed customer
- Dual Homed customer
- single multi-homed
- Dual multi-homed

single homed customer

single internet connection from ISP

we will never configure BGP in single homed customers.

we configure static default route in the direction of ISP and natting also

Dual homed customer

Two internet connection from same ISP

if we configure default route on both connection than it will not perform load-balancing

to utalize both of connection we use BGP in dual homed customer

Single multi-homed

In this we must have two ISP and customer must have atleast one connection from each ISP.

we can't configure default route so we to configure BGP

Dual multi homed

in this we must have atleast two isp and two link from each isp.

Types of msges in BGP

there is four types of msges in BGP, each type of BGP packet will travell over TCP. it is not possible to transmit any single msg of BGP without forming TCP connection.

1 OPEN msg

- open msg send only once when nbrship comes up. than after BGP will not send any open msg.
- open msg will be used by BGP to exchange initial configuration between two bgp peers which is require to form bgp neighborship

contents of open msg

- version 4
- type 1
- router-id
- own AS
- Hold down timer (180sec)

2 keep-alive

After exchanging open msgs successfully between both router, router will start to send keep-alive msgs periodically after every 60sec just to inform its neighbor is still awake.

if any router doesn't receive keep alive msg from its neighbor within 180sec (hold down time) than router will break down the neighborhood

type code 4

3 Update

update packets will be responsible to carry routes information. update packet having some components

Path Attributes: some criteria used by selection of best path.

prefix info/NLRI: network layer reachability information.

withdrawn routes: the prefix that goes down.

Note: if any router have multiple prefixes and all prefix attributes are same than router will generate a single update msg

4 Notification

if peering goes down than it will generate notification msg and it will contain the cause of problem.

FSM(finite state mechanism) of BGP / BGP neighborhood states

1 Idle: neighbor shutdown / waiting for start 3-way handshake.

2 Connect: try to form TCP connection, but wan't be confirmed

3 Active: TCP connection not completed, retrying to form tcp connection

4 Open-sent: TCP connection completed and sent open msg.

5 Open-confirm: exchanged open msg, perameter agreed, send keep-alive after 60sec

6 Established: peering completed, update is exchanged.

Types of neighbor in BGP

IBGP: Internal border gateway protocol

EBGP: external border gateway protocol

how to establish neighborship in bgp

```
router#debug ip tcp packet
```

```
router#debug ip bgp all
```

```
router(config)#router bgp 100
```

```
router(config-router)#neighbor 12.1.1.1 remote-as 200
```

```
router#show tcp brief
```

```
router#show ip bgp neighbor
```

```
router#show ip route
```

router#show ip bgp

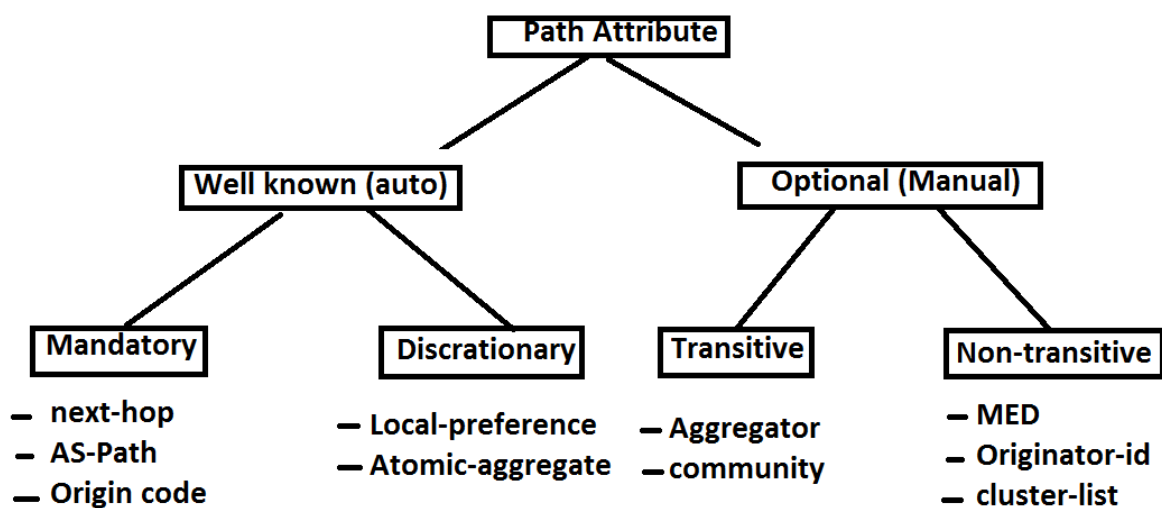
how to advertise prefix

router(config)#router bgp 100

router(config-router)#network 172.16.0.0 mask 255.255.0.0

note: we can advertise network but its prefix length should be match in its routing table connected route.

note: it will install only best route of bgp table into routing table.



Well known Mandatory

BGP will always carry these attributes with bgp update automatically

well know discretionary

sometimes bgp will carry these attribute with bgp update , sometime its not. it depends on nbr, its IBGP nbr or EBGP nbr

optional transitive

if any bgp neighbor receive any update with optional transitive attribute and that router dont support that attribute in that case it will not drop update, it will accept that update and also will forward it to another neighbor after removing that attribute.

optional non-transitive

if any bgp neighbor receive any update with optional non-transitive attribute and that router dont support that attribute in that case it will discard that incoming update.

how BGP define Best Path

1 Next-hop: reachability, if next-hop not reachable than it will not compare other path attribute, if any router receiving any prefix from two nbr and next-hop is reachable via both router, than bgp will not define best path based on next-hop.

2 weight:

- it is cisco proprietary and also it has local significance,
- bgp will not carry weight information with bgp update
- for self advertised prefixes it is by default 32768
- for all received prefixes by default it is 0
- higher weight will always be preferred.

3 Local preference

- bgp will always include local preference attribute in its outgoing bgp update but only for ibgp neighbor.
- by default local preference is 100 for internal prefixes
- higher will always be preferred
- if any router receive prefixes from ebgp nbr (0), and from IBGP nbr (100) in that case local preference will never compare

4 Self Advertised

- self advertised prefix will be preferred

5 AS Path

- AS- path list can have maximum 64 AS's
- router will define that path as best which is providing destination network after crossing less number of AS

6 Origin code

- i = if we are advertising route in bgp with network cmd
- E= if any route advertised in EGP.
- ? = if we redistributed anything in BGP by default origin code will be '?'
- i>E>?

7 MED (metric)

- default med is 0, lower will always be preferred
- if we redistribute any routing protocol into bgp, their metric will go along will bgp update in the form of med.
- some time metric will be 0 sometimes null.

8 external>internal

- if any prefix coming from two different nbr one is ebgp and one is ibgp then it will prefer external first

9 IGP cost to reach next-hop

- if igp configured background of bgp then bgp can use igp cost to reach next-hop address.
- lower cost will be preferred

10 oldest path (for ebgp neighbor only)

- if router receiving same prefix coming from two different EBGP neighbor then oldest path always be preferred

11 router-id of neighbor (for ibgp neighbor only)

- lower will always preferred
- if router receiving same prefix coming from two different IBGP nbr then lower router-id of neighbor will always be preferred

12 lower neighbor address

- if router receiving same prefix coming from two nbr which router id is same then lower ip address will be preferred

note: BGP support equal cost load-balancing but up to igp cost to reach next hop attribute must be same.

loop avoidance mechanism in BGP

BGP will always discard all these updates which are carrying its own AS inside their AS-path list.

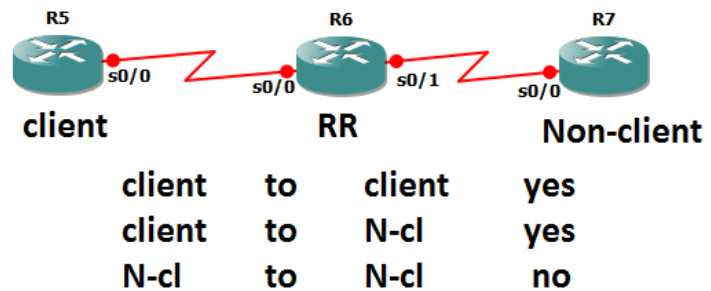
if a BGP router receives anything from one ibgp neighbor then it will never advertise that information toward any other ibgp nbr.

Rule of next-hop

- Whenever any router give update to its ebgp neighbor, it will advertise update after modifying their next-hop address.
- whenever any router give update to its ibgp neighbor, it will advertise update without modifying their next-hop address.

Route-Reflector

- RR server
- client
- non-client



how to configure neighbor

```
router(config-router)#neighbor 12.1.1.2 remote-as 100
```

how to advertise network in bgp

```
router(config-router)#network 10.1.1.0 mask 255.255.255.0
```

note: that prefix should be present in its local routing table with same prefix length.

connected check rule for EBGp

- According to BGP, EBGp neighbor are directly connected
- if ebgp neighbor are not directly connect in that case bgp will not initiate bgp neighborhood due to directly connected rule.
- to break this rule we have to disable connected check rule.
- if we change ttl value than directly connected check rule will get disabled automatically.

Note: open msg ttl value for ebgp neighbor is 1, and for ibgp neighbor ttl value is 255.

```
router(config)#router bgp 100

router(config-router)#neighbor 2.2.2.2 remote-as 200

router(config-router)#neighbor 2.2.2.2 update-source loopback 1

router(config-router)#neighbor 2.2.2.2 disable-connected-check

router(config-router)#neighbor 2.2.2.2 ebgp-multihop 2
```

how to advertise network in bgp

```
router(config-router)#network 10.1.1.0 mask 255.255.255.0

router(config-router)#network 192.168.101.0 (advertise class full prefix without mask)

router(config-router)#redistribute connected
```

how to check check that which route is advertising to neighbor router

```
router# show ip bgp neighbors 192.168.12.1 advertised-routes
```

```
router#clear ip bgp * in
```

how to configure authentication

```
router(config-router)#neighbor 1.1.1.1 password 5 cisco@123
```

how to change keep-alive interval and hold time

```
router(config-router)#timer bgp 50 100 (globly)

router(config-router)#neighbor 192.168.12.2 timers 50 100 (per neighbor)

note: bgp neighbors will negotiate lower hold timer
```

how to configure summarization

```
router(config-router)#aggregate-address 100.1.0.0 255.255.248.0 summary-only
```

we use path attribute for path manipulation for outgoing traffic (weight,local preference)

how to manipulate path with using weight for all prefix

```
router(config-router)#neighbor 192.168.13.3 weight 1
```

how to manipulate path selection with using weight for particular prefix

```
router(config)# access-list 1 permit 1.1.1.1 0.0.0.0
```

```
router(config)#route-map weight permit 10
```

```
router(config-route-map)#match ip address 1
```

```
router(config-route-map)#set weight 1
```

```
router(config)# route-map weight permit 20
```

```
router(config-router)# neighbor 192.168.13.3 route-map weight in
```

```
router# clear ip bgp * in
```

how to manipulate path selection with using local preference

```
router(config)#access-list 1 permit 1.1.1.1 0.0.0.0
```

```
router(config)#route-map lp permit 10
```

```
router(config-route-map)#match ip address 1
```

```
router(config-route-map)#set weight 1
```

```
router(config)#route-map lp permit 20
```

```
router(config-router)#neighbor 192.168.13.3 route-map lp in
```

how to shutdown any neighbor

```
router(config-router)#neighbor 192.168.12.2 shutdown
```

how to manipulate path selection with using AS-Path list

specially use this attribute to manipulate in bound traffic

```
router(config)#access-list 1 permit 1.1.1.1 0.0.0.0
```

```
router(config)#route-map as_path permit 10
```

```
router(config-route-map)#match ip address 1
```

```
router(config-route-map)#set as-path prepend 100 100
```

```
router(config)#route-map cisco permit 20
```

```
router(config-router)#neighbor 22.1.1.1 route-map as_path in
```

```
router#clear ip bgp *
```

how to manipulate path selection with using origin code

this attribute is also used to manipulate in bound traffic

```
router1(config)#access-list 1 permit 1.1.1.1 0.0.0.0
```

```
router1(config)#route-map as_path permit 10
```

```
router1(config-route-map)#match ip address 1
```

```
router1(config-route-map)#set origin incomplete
```

```
router1(config)#route-map cisco permit 20
```

```
router1(config-router)#neighbor 192.168.12.2 route-map as_path in
```

```
router1#clear ip bgp *
```

how to manipulate path selection with using med

this is also used for inbound traffic

```
router1(config)#access-list 1 permit 1.1.1.1 0.0.0.0
```

```
router1(config)#route-map med permit 10
```

```
router1(config-route-map)#match ip address 1
```

```
router1(config-route-map)#set metric 1
```

```
router1(config)#route-map cisco permit 20
```

```
router1(config-router)#neighbor 192.168.12.2 route-map med in
```

```
router1#clear ip bgp *
```

note: if any prefix is receiving from 2 different neighbor and both neighbor AS is same than it will compare med. if any prefix is receiving from 2 different neighbor and both neighbor AS is different than it will not compare med (bydefault)

```
router(config-router)# bgp always-compare-med
```

note: we cant manipulate path selection with using path-attribute external>internal.

redistribute eigrp into bgp

```
router(config-router)# redistribute eigrp 100
```

```
redistribute bgp into ospf
```

```
router(config-router)#redistribute bgp 300 subnets
```

```
redistribute ospf into bgp
```

```
router(config-router)#redistribute ospf 100 / match internal external
```

```
redistribute bgp into eigrp
```

```
router(config-router)#redistribute bgp 200 metric 1544 0 255 1 1500
```

note: bydefault bgp will not redistribute internal route.

```
router(config-router)#bgp redistribute-internal
```

route filtering in bgp

standard ACL

prefix-list

route-map

note: we can't call route-map in distribute-list

route filtering with using standard acl

```
router2(config)# access-list 1 deny 40.1.1.1 0.0.0.0
```

```
router2(config)# access-list 1 permit any any
```

```
router2(config-router)# distribute-list 1 in
```

or

```
router2(config-router)# neighbor 192.168.2.2 distribute-list 1 in
```

route filtering with using prefix-list

```
router(config)# ip prefix-list test deny 40.1.0.0/22 ge 32 le 32
```

```
router(config)# ip prefix-list test permit 0.0.0.0/0 le 32
```

```
router(config-router)# neighbor 192.168.2.2 distribute-list prefix-list test in
```

route filtering with using route-map

route

ipv6 addressing

it is a 128 bits address and divided into 8 blocks, in each block there is 16 bits

it will be represented in hexadecimal form

ipv6 address is divided into two parts, first part known as network prefix, and second part is host-id part.

types of address in ipv6 = 1 unicast 2 multicast

unicast: unique local address (fc00::/7), global unicast address (2000::/3), link local add (fe80::/10)

multicast: ff00::/8

loopback address ::1

link-local address

we will never required to allocate link-local address manually

device will automatically calculate these link-local address by itself

link local address can't across its broadcast domain

we can communicate within the network with using this address.

NO ARP

NO need of dhcp server (serverless auto configuration)

ND protocol

this protocol will use icmpv6 to get the mac address of dest/next-hop, serverless auto conf, router-discovery, prefix discovery, duplicate address

DAD duplicate address detection

there is no gratuitous arp, to detect duplicate ip address it uses DAD, and this dad will be also performed with the help of icmpv6.

types of msgs in ipv6

router solicitation:

multicast by a node when it wants routers to send a router advertisement immediately instead of waiting for advertisement the next scheduled. it send router solicitation to learn about config parameters and about the existence of routers on the link.

router advertisement

sent periodically or in response of a solicitation. routers advertise their presence, as well as provide information necessary for a node to configure itself. (gateway info, prefix info)

neighbor solicitation

it is like arp req msg, to get layer2 address of neighbor, determine whether the neighbor is still reachable via layer2 address, to check duplicate address.

neighbor advertisement

it is like ARP response msg, to give layer 2 address.

redirect

if there is two different prefix and they will communicate via router so after redirect msg they will communicate directly

ipv4 224.0.0.1 for all node, 224.0.0.2 for all routers only

ipv6 ff02::1 for all node, ff02::2 for all routers only.

rip ff02::9

ospf ff02::5/6

eigrp ff02::A

for stateless auto config

router(config-if)# ipv6 address autoconfig

router(config-if)# ipv6 address autoconfig default (to receive default route automatically)

RIP:ex

we can use offset-list in rip (config-if)#ipv6 rip test metric-offset 3)

rip supports only prefix list for route filtering

BGP

router(config)#ipv6 unicast-routing

router(config)#router bgp 100

router(config-router)# bgp router-id 1.1.1.1

router(config-router)#neighbor 2000:192:168:1::2 remote-as 200

router(config-router)# no bgp default ipv4-unicast

router(config-router)#address-family ipv6 unicast

router(config-router)#neighbor 2000:192:168:1::2 activate

router#show ipv6 unicast summary

GRE: Generic router encapsulation

it is a layer 3 protocol

it uses protocol number 47

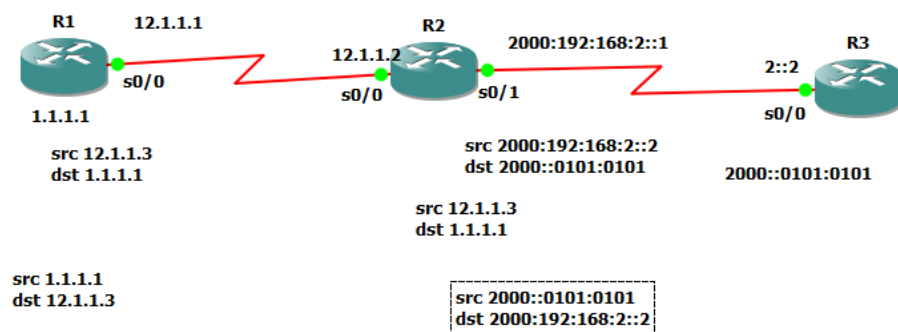
it will add 24byte long extra information (external ip header 20bytes, type and flag 4bytes)

compatability between ipv4 and ipv6

Dual stacking

ipv6 tunnel

NAT-PT



2000::/96 1.1.1.1 00000001. 00000001. 00000001. 00000001.
0 1 0 1 0 1 0 1

2000::0101:0101

```
r2(config)#ip route 1.1.1.0 255.255.255.0 serial 0/0
```

```
r3(config)# ipv6 route ::/0 serial 0/0
```

```
r2(config-if)# ipv6 nat (serial 0/0)
r2(config-if)#ipv6 nat (serial 0/1)
r2(config)# ipv6 nat prefix 2000::/96
r2(config)#ipv6 nat v6v4 source 2000:192:168:2:2 12.1.1.3

r2(config)# ipv6 nat v4v6 source 1.1.1.1 2000::0101:0101
```

GRE Generic router encapsulation

layer 3 protocol

protocol number is 47

will directly interact with ip itself

it will add 24byte long extra information

24byte= 20byte for external ip header, 4byte will be for type and flags field.

DMVPN

NHS: ROUTER(CONFIG)# interface tunnel 1

```
router(config-if)#ip address 192.168.1.1 255.255.255.0
```

```
router(config-if)# tunnel source serial 1/0 / 11.1.1.1
```

```
router(config-if)#tunnel mode gre multipoint
```

```
router(config-if)#ip nhrp network-id 1 (any number)
```

```
router(config-if)#ip nhrp authentication cisco123
```

NHC:

```
router(config)#interface tunnel 1
```

```
router(config-if)#ip address 192.168.1.2 255.255.255.0
```

```
router(config-if)#tunnel source serial 1/0
```

```
rotuer(config-if)#tunnel destination (ip address of nhs)
```

```
rotuer(config-if)#ip nhrp network-id 10
```

```
rotuer(config-if)#ip nhrp authentication cisco123
```

rotuer(config-if)#ip nhrp map 192.168.1.1 (nhs tunnel ip address.) 11.1.1.10 (public ip of nhs)

```
rotuer(config-if)#ip nhrp nhs 192.168.1.1
```

```
rotuer#show ip nhrp
```

after configuring eigrp

```
nhs(config-if)# ip nhrp map multicast dynamic (tunnel interface)
```

```
nhc(config-if)#ip nhrp map multicast 11.1.1.1 (public ip of nhs)
```

```
nhc(config-if)#shutdown / no shutdown.
```

phase2 tunnel 1 = tunnel mode gre multipoint

phase3 tunnel 1 (nhs) = ip nhrp redirect

tunnel1 (nhc)= ip nhrp shortcut

Site to Site VPN / IP Sec VPN

Data integrity (without manipulated)

data encryption

authentication

Anti-replay

there are two phases

phase1 = ISAKmp (internet security association key management protocol)

phase2= IPsec

mpls

Router(config)#mpls ldp router-id lo 0

router(config)#mpls label protocol ldp/tdp (bydefault ldp)

router(config-if)#mpls ip (to enable mpls on any interface)

router#show mpls ldp binding (to check LIB table)

router#show mpls forwarding table (to check LFIB Table)

AAA (authorization)

priviledge level (default 0,1,15)

```
r(config)#username cisco privilege 5 password cisco
```

```
r(config)#username test privilege 15 password test
```

```
r(config-line)#login local
```

login with cisco user

it can run only limited commands.

now login with test user which has privilege 15 and give the privilege to user cisco.

```
r(config)#privilege exec level 5 configure terminal
```

```
r(config)#privilege configure level 5 interface
```

```
r(config)#privelege interface level 5 ip address
```

```
r(config)#privilege interface level 5 shutdown
```

```
r(config)#privilege interface level 5 no shutdown
```

```
r# show privilege
```

syslog server

r(config)#login console ?

r(config)#login monitor ?

r(config)#logging host 192.168.1.1 transport udp port 656

r(config)#logging source-interface loopback 1

r(config)#logging rate-limit console all 2

snmp server

r(config)#snmp-server host 192.168.1.1 cisco@123

r(config)#snmp-server community cisco@123 rw

r(config)#snmp-server enable traps

chck in running config

r(config-if)#no snmp trap link-status